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Monitoring of heart rate, blood oxygen saturation, and blood pressure using a smartphone

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Highlights

- Using single smartphone for SpO_2, HR and BP estimation.
- Development of unique Android app; it provides real-time audio-visual feedback.
- Complex study - 65 signals, 22 people, 13 smartphone models.
- Improved calculation of SpO_2 without the need of smartphone quantum efficiency.
- Errors: 1.1%, 1.4 bpm, (5.1 and 7.5) mmHg for SpO_2, HR, and BP estimation, resp.

Abstract

Smartphones can be used as multifunctional devices for instant monitoring of the users' health. In this study, we proposed methods for the estimation of heart rate (HR), blood oxygen saturation (SpO_2), and blood pressure (BP). Two smartphone sensors were used for data recording: the rear camera and the microphone. Our approach for HR and SpO_2 estimation was based on creating a photoplethysmogram (PPG) from the camera data. SpO_2 estimation was improved and the smartphone quantum efficiency did not have to be known. BP was estimated using the pulse transit time value calculated from the PPG and phonocardiogram (PCG) recorded using the microphone. Fifty records (for HR and SpO_2) and 15 records (for BP) were sensed and used for training and testing purposes. An Android™ application was developed to provide audio-visual feedback to the measured subject, to indicate whether the placement of the smartphone was appropriate to provide signals of sufficient quality for BP estimation. The mean absolute errors (MAE) were 1.1% for SpO_2 and 1.4 beats per minute (bpm) for HR estimation using the testing dataset. MAE of systolic and diastolic BP were 5.1 mmHg and 7.5 mmHg, respectively. The possibility of using the smartphone as a fast alternative to conventional and specialized devices for SpO_2, HR, and BP estimation was statistically proven.

Keywords

Heart rate; Blood oxygen saturation; Blood pressure; Photoplethysmogram; Health monitoring; Smartphone; Android application

Introduction

It has been estimated that there are 2.71 billion smartphone users worldwide in 2019 [1], with this number increasing yearly. The sensors integrated into smartphones, and their computational power enables wide usage in many areas of daily life, including health and activity monitoring. Parameters such as heart rate (HR) [[2], [3], [4], [5], [6],29,31,34,36,[38], [39], [40], [41]], blood oxygen saturation (SpO_2) [7,8,29,30], blood pressure (BP) [[9], [10], [11], [12], [13],37], number of walking steps [14], and type of activity [15] can be measured using a smartphone. Smartphone health monitoring does not replace certified medical devices, but it can be used as an approximate alternative. Its advantages are multifunctionality, mobility, and non-invasiveness.

This study deals with the possibilities of HR, SpO_2, and BP monitoring using a smartphone. As conventional techniques and gold standards, ECG, gas chromatography (GC) [8], and mercury sphygmomanometry are used, respectively. These measurements are reliable, assuming the measurement is provided correctly by a trained physician. Moreover, for GC, invasive blood collection is necessary. For home monitoring, devices such as pulse oximeter (PO) or oscillometric sphygmomanometer, which are more user-friendly but usually less accurate, exist.

HR can be estimated using one of the following smartphone sensors: rear camera [[2], [3], [4], [5], [6],34,36], front camera [31,40], accelerometer [38,41], microphone [39], or sensors dedicated to HR and SpO_2 measurement [29]. The data from the rear camera are usually recorded from the finger, processed into the photoplethysmogram (PPG) waveform, and further analyzed to calculate HR [[2], [3], [4], [5], [6],34,36]. The front camera is used for the estimation of HR from the face. The data are processed similarly to the data from the rear camera, the PPG is computed, and further analysis is provided [31,40]. Using the smartphone's built-in accelerometer, the mechanical activity of the heart can be measured (the seismocardiogram or ballistocardiogram) [38,41]. These signals are filtered, then the principal component analysis (PCA) [38] or linear power, quadratic spline decomposition and continuous wavelet transform [41] are performed, and finally, the peaks are searched. The microphone-based approach uses envelope calculation and peak detection [3]. Tayfur et al. [29] use sensors built-into the Samsung Galaxy S8 for HR and SpO_2 estimation along with the Samsung Health application for evaluation of the accuracy by comparisons with a vital signs monitor (VSM).

The SpO_2 level is a ratio between hemoglobin carrying oxygen and the total amount of hemoglobin in the blood [16]. Normal values of SpO_2 are 96–98% [16] or even up to 100%. It is usually measured by the PO. The measurement principle is similar to that with the smartphone. Both are based on photoplethysmography (PPG), which detects blood volume changes. The smartphone-based solution is simpler. Using the light-emitting diode (LED) light near to the camera is similar to the reflectance type measurement of the PO [17]. It approximates two light sources of different wavelengths by the red and green RGB components of the recorded video [8]. The SpO_2 estimation using the smartphone is based on PPG features extraction and the use of mathematical equations [7,8]. Ding et al. [30] also calculate PPG and then use support vector machine (SVM)-based motion detection, feature extraction, and convolutional neural networks.

Arterial BP is usually measured in its two extremes – the highest (systolic) and the lowest (diastolic) [18]. Physiological values oscillate around 120/70 mmHg [18]. The most promising methods for BP estimation using the smartphone are based on the measurement of pulse (or vascular) transit time (PTT) [9,10]. PTT is defined as the time necessary for the pulse wave (blood) to travel from the heart to a periphery (fingertip). This feature is a function of the elasticity of the blood vessels and, thus, also with BP (lower elasticity means higher BP) [10]. PTT can be calculated from two signals measured by sensors placed on two different parts of the body (close to the heart and on the periphery). The combination of signals can be i) ECG

imaging) [12], b) phonocardiogram (PCG) and PPG [9,10], and c) seismocardiogram and PPG [23]. BP is then calculated using equations [9]. Some methods need only one PPG for BP estimation [24,25,11,35,37,32]. These methods are based on feature extraction and the use of machine learning methods [24,25,35,37], cardiac output and total peripheral resistance estimation [11], and morphology and frequency analysis of PPG [32]. Dey [25] developed the Android application InstaBP for BP monitoring using a PPG sensor integrated into the flagship Samsung smartphone models. The limitation of the InstaBP application is that it does not work in all countries (e.g., the Czech Republic; tested on the Samsung Galaxy S7). The advantage of smartphone-based methods is the fact that they can evaluate BP from each beat, which is not possible in the case of cuff-based measurement [23]. Chandrasekhar et al. [26] recently introduced the innovative oscillometric finger pressing method. The authors utilized the iPhone X with a strain gauge array under the screen (3D Touch), which measures the pressure of fingertip placement. The second sensor used is a front camera, which measures blood volume oscillations in the finger. For the recording, the authors developed an iOS application which leads the user while pressing the touch screen of the smartphone, because it is necessary to change the fingertip pressure during the measurement time. The BP estimation is similar to cuff-based oscillometric methods and is done offline. The technique of Luo et al. [31] records the video of the face, extracts 126 features and adds 29 meta-features, and uses the multilayer perceptron machine learning method. Peng et al. [33] use only the PCG recorded by the smartphone microphone to estimate BP. The principle is based on the extraction of heart-sounds features and the use of the SVM [33].

The remainder of this paper is organized as follows. The procedure of data recording and the proposed methods for SpO_2, HR, and BP estimation are described in Section 2. In Section 3, the results are presented. In Section 4, the achieved results are discussed and compared with those of other authors, and the limitations of the study are described. In Section 5, the conclusions drawn from the study are presented.

Section snippets

Data recording and the developed Android application

To develop the algorithm for HR and SpO_2 estimation, the training database was created. It includes 30 records, each approximately 20 s in length, from eight people. Each record (video) was sensed using the smartphone rear camera with illuminated LED light while the index finger was placed on both, as illustrated in Fig. 1. Three smartphones were used – the Honor 7 Lite, the Apple iPhone SE, and the Lenovo Vibe S1. The framerate used was 30 frames per second (fps). The algorithm was developed …

Results

Table 1 presents the results for SpO_2 and HR estimated in the training dataset (calculated as a mean from the whole signal). This table shows both the reference values of SpO_2 and HR as well as the calculated values, and also their differences (errors). The training set mean absolute error (MAE) of SpO_2 and HR are 1.1% and 1.3 bpm, respectively. The maximum errors (MAX) for SpO_2 and HR are 4% and 4 bpm, respectively. The errors were also calculated according to Eq. (8) [3] – denoted as *error2*. …

Discussion

Our algorithm for SpO_2 estimation does not account for quantum efficiency (QE), as some other algorithms do [7]. QE can be different for different smartphones (cameras) and is usually not publicly available. Moreover, the SpO_2 is calculated with an error of 1.1% on both training and testing datasets without the use of QE.

Conclusions

The contributions of this study are derived from the use of only a single smartphone for SpO_2, HR, and BP estimation; the development of a unique Android application; the complexity of the study – it involves SpO_2, HR, and BP monitoring; the use of 65 signals from 22 people using 13 smartphone models; and the improved calculation of SpO_2 without the need to know the smartphone QE parameter. It was statistically proven that it is possible to use the smartphone as an approximate alternative to …

CRediT authorship contribution statement

Andrea Nemcova: Conceptualization, Investigation, Methodology, Visualization, Writing - original draft, Writing - review & editing. **Ivana Jordanova:** Conceptualization, Formal analysis, Investigation, Methodology, Writing - review & editing. **Martin Varecka:** Conceptualization, Formal analysis, Investigation, Methodology, Software, Writing - review & editing. **Radovan Smisek:** Conceptualization, Methodology, Writing - original draft, Writing - review & editing. **Lucie Marsanova:** Conceptualization, …

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. …

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Abstract

Introduction

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